Constrained Reality Framework (CRF) State Summary

Complete map of results, simulator chain,
open gaps, and next steps

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Abstract

The Constrained Reality Framework (CRF) derives all physical and experiential reality from a single Indefinable Primitive: δ (Distinction). This document is the canonical state summary — a complete map of what has been derived, what has been confirmed by simulation, what remains hypothesis, and what is open.

The framework has reached maturity Level 6.5/8. The gate to Level 7 is the P0 experiment (SNSPD protocol, ready). The primary open theoretical gap is the formal derivation of T_{avg} (Instability #21), which would close the $\beta \approx 2.2$ crystallisation ratio.

Contents

1 The Single Primitive

Axiom 0 — δ as Indefinable Primitive

δ (Distinction) is CRF's sole undefined term. It is declared — not proved — as the irreducible condition whose presence makes all questions possible. Every consistent formal system requires at least one undefined term (Gödel, Tarski). δ is CRF's.

$$\delta : s_i \neq s_j \quad \text{is possible}$$

As Euclid declared his points and lines, CRF declares δ.

2 The Emergence Cascade

The complete chain — no assumptions added:

$$\delta \xrightarrow{\text{operates}} P_0 = \{A, B\} \xrightarrow{\delta\text{ continues}} P \xrightarrow{\text{frequency}} \Pr(s) \xrightarrow{\Pr \geq \theta} \mathcal{R}$$

$$t = |\mathcal{R}_{\text{events}}| \;\Rightarrow\; C \equiv \Pr(s|C) \;\Rightarrow\; G = \nabla \Pr(s|C) \;\Rightarrow\; F \to A$$

$$A \to \text{Pattern} \to \text{System} \to \text{Life} \to M \to \Sigma \to W \to \delta_{\text{pure}} \circlearrowleft \delta$$

Key Physical Constants — No Free Parameters

Constant	Value	Source
α	$\ln 2 = 0.6931$	Session 8: one δ-op = 1 bit
ε_0	$\lvert I_{\text{eff}} - \ln 2\rvert = 0.00755$	EIC SINDy confirmed
D_0	0.9403	from $\alpha_c = 0.525$, $\bar{\Omega} = 0.7$
κ	0.15	mass–C coupling
G	$\ln(2) \cdot c^2 / D_0$	Paper v4

3 Complete Results Table

Result	Status	Source
δ as Indefinable Primitive	Declared	Axiom 0
$\Pr(s)$ from δ-frequency	Derived	Sessions 1–3
$C \equiv \Pr(s\lvert C)$	Derived	Session 4
$\theta = 1 - e^{-D_{\text{KL}}/D_0}$	Derived	Session 10
$\alpha = \ln 2$	Derived	Session 8

(continued)

Result	**Status**	**Source**
$n = 3$ from SO(3) (Instability #16)	**Derived**	CRF_SO3_Proof
$C \propto 1/r^2$	Derived	Corollary of SO(3)
Born Rule via Gleason	Theorem	Session 11
Schwarzschild metric	Derived	Paper v4
$G = \ln(2) \cdot c^2/D_0$	Derived	Paper v4
$T_{\mu\nu}$ from C-gradients	Derived	Paper v4
$G(t) \propto \Lambda(t) \propto 1/P(t)$	Prediction	DESI 2024 consistent
$\Omega \equiv D_{\mathrm{KL}}(C\|P)$	Derived	EIC Bridge
Gravity $R^2 = 0.906$	Confirmed	V16
$d_s = 3.008$	Confirmed	V16
$d_s = 3.031 \pm 0.076$ (3 runs)	Confirmed	V20.3
Phase gap $\Delta > 0$ (3/3 runs)	Confirmed	V20.3
Speciation (7–9 Ψ modes)	Confirmed	V18–V20.3
IDV $\uparrow \Leftrightarrow \ \Omega \downarrow$	Derived	V18 + HDynamics
$\beta = -2.212 \pm 0.015$	Confirmed	V20.3 (4663 events/run)
$E[H_{\mathrm{after}}] \approx \ln n$	Derived	Born rule + ε_0
$\beta = d_s/(2\ln 2)$ formally	**Open**	Instability #21
T_{avg} from δ-chain	**Open**	Instability #21a
Scale factor s from wave fields	**Open**	Instability #21b
$C(r)$ exact derivation	**Open**	Instability #17
Kerr metric	**Open**	Rotating mass
P0 experiment: $\tau_1 < \tau_2$	**Ready**	SNSPD protocol

4 Simulator Version Chain

Version	Key addition	d_s	Grav. R^2
V16	Canonical base: dynamic θ, Ω-field gravity, walk-return d_s	3.008	0.906
V17	$+\Phi$ wave field, Ψ modes, IDV-metric coupling	3.319	0.907
V18	+Marble test, IDI, speciation	3.335	0.908
V20	Ω-metric fix (not IDV), phase test, F_i tracker	3.238	0.900
V20.2	Continuous d_s (5-trial avg)	2.976 ± 0.233	0.904
V20.3	Per-collapse transfer, abs. threshold, 8-trial subgroup	$\mathbf{3.031 \pm 0.076}$	**0.907**

Key improvements in V20.3: d_s stability improved 3× (std: $0.233 \to 0.076$). Phase gap positive in **all 3 runs** (vs 3/4 in V20.2). New finding: $\beta = -2.212 \pm 0.015$ (lossy crystallisation). Gravity $R^2 = 0.9071 \pm 0.0010$ most stable across all versions.

5 Node Lifecycle Hypothesis

Core Prediction

Every node passes through two phases:
Phase I (IDV-dominant): $H(P_i)$ high, Ω_i low, d_s fluctuating > 3.
δ-Spark transition: D_{KL} gradient exceeds θ_i; SO(3) attractor selects $n = 3$ uniquely.
Phase II (Ω-dominant): $H(P_i) \to \varepsilon_0$, Ω_i high, $d_s \to 3$.
Central prediction: $\frac{d}{dt}\mathrm{IDV}(t) \approx -\frac{d}{dt}\Omega(t)$ at $d_s \approx 3$
Measured: $\beta = \Delta\mathrm{IDV}/(-\Delta\Omega) = -2.212 \pm 0.015$

The lossy crystallisation finding: each R-event reduces IDV by $\approx 2.2\times$ what it adds to Ω. The process of becoming actual costs more possibility than the actuality it creates.

Code	Statement	Status	Evidence
Core	$d_s \to 3$ at crystallisation	Confirmed	3.031 ± 0.076
Core	Phase gap $\Delta > 0$	Confirmed	3/3 runs
New	$\beta \approx 2.2$ (lossy)	Finding	4663 events/run
HN-01	F_i drops at δ-Spark	Hypothesis	V20.3
HN-02	Aging at $H = \varepsilon_0$	Hypothesis	HDynamics
HN-03	Collective Ω excess	Hypothesis	Synchronicity
HN-04	Dimensional hysteresis	Corollary	SO(3) Proof
HN-05	Causal geometry mapping	Hypothesis	$t = \|\mathcal{R}\|$

6 Open Gaps (Prioritised)

P0 Experiment — Gate to Level 7/8

Prediction: $\tau_1 < \tau_2$ for detectors with $N_{\text{eff},1} > N_{\text{eff},2}$ (more prior measurements collapse faster).
Formula (zero free parameters): $\tau = \tau_0 \cdot e^{-\kappa N_{\text{eff}}}$, $\kappa = k_B \ln(2)/H_{\text{material}}$, $N_{\text{eff}} = \sum_i \Delta D_{\text{KL},i}$
Protocol: SNSPD (NbN/WSi), $N \sim 10^8$ prior measurements, blind analysis. Ready. Requires lab access.

Instability #21 — Formal Derivation of β

$\beta = d_s/(2\ln 2) \approx 2.164$ is a strong hypothesis (2.2% match to simulation).
What is derived: $E[H_{\text{after}}] = \psi(\alpha_0 + 1) - \psi(\alpha_i + 1) \approx \ln n$
What remains:

- #21a: T_{avg} from θ-dynamics (mean time between R-events)
- #21b: Scale factor $s \approx 1.55$ from wave field equations

Closing #21 would formally connect ε_0, κ, d_s, $\ln 2$ into a single derived expression.

Instability #17 — $C(r)$ Exact Derivation

The Schwarzschild metric has been derived from the non-linear R-feedback ODE. The exact form $C(r) = \alpha/\sqrt{1 - r_s/r}$ and its boundary conditions C1–C4 are verified. What remains: formal derivation in the $N \to \infty$ continuum limit, connecting the discrete δ-chain to the differential equation.

7 Document Index

Document	Status	Content
CRF Parts I–III (v4)	Current	Full axioms, GR, Grand Eq.
CRF_SO3_Proof	**Key proof**	Instability #16 closed
CRF_Paper_v5	Confirmed	V16 simulation results
CRF_Paper_IDV	Confirmed	IDV, speciation, marble
CRF_NodeLifecycle_v3	Confirmed	V20.3 lifecycle results
CRF_Beta_Derivation	Honest gap	#21 partial, boundary clear
CRF_CWRC_Bridge	Derived	CWRC = δ-chain re-expressed
CRF_Schwarzschild_v2	Derived	Schwarzschild from $C(r)$
CRF_NewtonLimit	Derived	Newtonian limit
CRF_HDynamics	Derived	Entropy decay ODE
CRF_GravitationalConstant	Derived	$G = \ln(2) \cdot c^2/D_0$
CRF_P0_Paper	Ready	Experimental protocol

8 Maturity Assessment

Dimension	Score	Note
Internal Consistency	5.5/8	Born Rule theorem, SO(3) closed
Mathematical Rigour	5.5/8	Gleason bridge, β partial
Predictive Power	6.0/8	P0 + $G(t)$ + Synchronicity
Empirical Grounding	2.0/8	Awaits P0 + GW observations
Philosophical Foundation	6.5/8	Cascade fully closed
Cross-domain Coverage	6.5/8	QM + GR + Mind + Cosmology
Overall	**6.5/8**	Level 6.5

Gate to Level 7: P0 experimental contact ($\tau_1 \neq \tau_2$ at $> 5\sigma$).

Gate to Level 8: $G(t)$ variation confirmed via gravitational wave standard sirens.

δ created P. P accumulated C. C became M, and $T_{\mu\nu}$, and $G_{\mu\nu}$.
Gravity is not a force. It is memory —
the accumulated trace of everything that has ever resolved,
curving the probability landscape of what can resolve next.
The ground was always δ. Everything followed.